

Passenger Price Sensitivity

Simple Implementation Steps — Clean Reference Version

Purpose: Calculate passenger price sensitivity from the Jan–Mar 2026 historical training data for each Time × Weather condition, then calculate the mean and standard deviation.

Step 1 - Use the complete historical training dataset

Use the Jan–Mar 2026 training data, covering all origin–destination pairs, all 24 hourly slots, and all observed weather types.

Step 2 - Determine the base/reference values

Using the complete historical dataset, determine the base/reference distance L_{base} and base/reference price P_{base} . These are then used consistently for the subsequent calculations.

Step 3 - Select one Time × Weather pair

Example: **08:00 × Clear**. Extract all historical trips belonging to that condition. For every trip, retain its observed distance L_i and price P_i .

Step 4 - Calculate relative price change

For every trip i :

$$\Delta P_i = (P_i - P_{\text{base}}) / P_{\text{base}}$$

Step 5 - Calculate relative distance change

For every trip i :

$$\Delta L_i = (L_i - L_{\text{base}}) / L_{\text{base}}$$

Step 6 - Calculate individual sensitivity

For every trip:

$$\varepsilon_i = \Delta L_i / \Delta P_i$$

Equivalent form:

$$\varepsilon_i = [(L_i - L_{\text{base}}) / L_{\text{base}}] \div [(P_i - P_{\text{base}}) / P_{\text{base}}]$$

Step 7 - Create the sensitivity array

Store all ε_i values for the selected Time × Weather pair. This is the historical sensitivity array for that condition.

Step 8 - Calculate mean and standard deviation

From the sensitivity array calculate:

$$\text{Mean } \varepsilon = \text{mean}(\varepsilon_i)$$

$$\text{SD } \varepsilon = \text{standard deviation}(\varepsilon_i)$$

Step 9 - Repeat for all Time × Weather pairs

Repeat Steps 3–8 for every required hourly slot and weather type. Record the sample count, mean ε , and SD ε for each condition.

Step 10 - Reduce weather types into 3 classes

After the Time × Weather calculations, reduce the observed weather types into **3 weather classes**, as specified in the project procedure. The exact grouping method should follow the agreed project method.

Final Output Table

Hour	Weather	N	Mean ε	SD ε
00:00	Clear	...	...	...
00:00	Rain	...	...	...
00:00	Snow	...	...	...
01:00	Clear	...	...	...
...	...	...	...	...
23:00	Snow	...	...	...

Overall Calculation Flow

Jan–Mar data → Determine L_{base} and P_{base} → Select Time × Weather → Relative ΔP → Relative ΔL → ε
 $= \Delta L / \Delta P \rightarrow \varepsilon$ array → Mean + SD → Repeat for all groups → 3 weather classes

Important: Both ΔP and ΔL are relative changes. Therefore the denominator is $(P_i - P_{base})/P_{base}$, not the raw price difference alone.